08. Model Evaluation

Interpret the top-five retrieval example and define proper recommender metrics.

Project: Movie Recommender System

Source archive supplied in this conversation: Movie-recommender-system-main.zip | Notebook: notebook.ipynb | App: app.py

Observed example

The notebook calls recommend('Gandhi') and prints five titles: Gandhi, My Father; The Wind That Shakes the Barley; A Passage to India; Guiana 1838; Ramanujan. This is a qualitative example, not a measured evaluation result.

How recommendation works

- Find the row index of the selected title in the movie table.
- Read that row's similarity scores against all movies.
- Enumerate scores and sort in descending order.
- Skip the first entry (the selected movie itself, whose self-similarity is normally 1).
- Return the next five titles.

Ranking metrics to know

Metric	Meaning
Precision@K	Fraction of top-K recommendations judged relevant
Recall@K	Fraction of all relevant items retrieved in top K
MAP@K	Average precision across ranked relevant items
NDCG@K	Rewards relevant items higher when they appear earlier
Coverage	Share of catalogue items that can be recommended
Intra-list diversity	How different the recommended items are from each other
Novelty	Whether recommendations expose users to less obvious items

Evaluation limitations

- No labelled relevance judgments or user feedback are present.
- No offline ranking metrics are computed.
- No baseline comparison is reported.
- No diversity, novelty, coverage, popularity-bias or long-tail analysis.
- The nearest-neighbour list can be dominated by shared cast, director or generic plot terms.
- Sorting all movies for every query is O(N log N) after similarity scores are available; precomputed full matrix storage is $O(N^2)$.

Interview-ready answer

“I validated the prototype qualitatively by checking the nearest titles for a sample seed movie. The repository does not contain user feedback or relevance labels, so I cannot claim measured precision or recall. For a rigorous evaluation I would collect interactions or human relevance judgments and report Precision@K, Recall@K, NDCG@K, coverage and diversity against simple baselines.”